Example

- What is the coefficient of $x^{12}y^{13}$ in the expansion of $(2x-3y)^{25}$?

Solution:

By the Binomial theorem the term involving $x^{12}y^{13}$ is $\binom{25}{13}(2x)^{12}(-3y)^{13}$. Therefore, the coefficient of $x^{12}y^{13}$ is

$$-\binom{25}{13}2^{12}3^{13}$$